

UNIT 1

SETS



SET

In mathematics collection of things is called set.

A set is a collection of 'well defined' and 'distinct' objects.

'Well defined' means a set must have some specific property.

'Distinct' means different objects.

Notation of a set and its objects/elements:

The objects of the set are called elements or members of the set. The elements of the set are listed within brackets $\{ \}$ and separated by commas.

The symbol ' $\in$ ' is used to show the membership to a set and read as "belongs to". ' $\notin$ ' means "does not belong" to a set.

Example 1:**i. $A = \{ 3, 5, 7, 9 \}$**

The members of set A are 3, 5, 7 and 9.

As 3 belongs to set A so, symbolically we write as $3 \in A$

As 4 does not belong to Set A.

So, symbolically we write as $4 \notin A$

ii. $C = \{ a, b, c \}$

The elements of set C are a, b and c.

Here, $a \in C$, but $p \notin C$

TABULAR FORM OF A SET

The elements of a set that are listed within $\{ \}$ and separated by commas are called tabular form.

Points to remember

- The name of a set is denoted by capital letter of English alphabet.
- Every element of a set is separated by a comma ',' in tabular form.
- All the members of set are enclosed in brackets $\{ \}$ in tabular form.
- The order of writing elements does not matter in a set.
- Elements of set do not repeat itself.

Example 1: Set of first five small letters of the English alphabets.
In tabular form we write as $\{a, b, c, d, e\}$

Example 2: Set of first six natural numbers
In tabular form we write as $\{1, 2, 3, 4, 5, 6\}$

Example 3: Set of first three even numbers.
In tabular form we write as $\{2, 4, 6\}$

Example 4: Set of first five names of month.
In tabular form we write as $\{\text{January, February, March, April, May}\}$

Example 5: Explain, why are the following not sets:

1. $X = \{2, 3, 4, 5, 5, 6, 7\}$
2. $B =$ Collection of brilliant students

Solution:

1. $X = \{2, 3, 4, 5, 5, 6, 7\}$
This is not set because the element 5 is repeated
2. $B =$ Collection of brilliant students
This is not set because it cannot be decided that which students are brilliant.

DESCRIPTIVE FORM

The descriptive form of a set is a way to represent a set using words or symbols to describe its elements.

Consider the following sets:

A = Set of students of class VII of your school

B = Set of first ten natural numbers

C = Set of first ten whole numbers

D = Set of geometrical instruments in a box

EXERCISE 1.1

Q1. Write the given sets in tabular form.

Set	Tabular Form
(i) Set of first five English alphabets.	$\{a, b, c, d, e\}$
(ii) Set of all factors of 12	
(iii) Set of names of days of a week which start from "F"	
(iv) Set of first five natural numbers	

Q2. Tick which of the following are sets.

i. Collection of the names of Prime Ministers of Pakistan.

☐

ii. Collection of names of captains of cricket teams of Pakistan.

☐

iii. Collection of delicious desserts.

☐

iv. Collection of brilliant students in your class.

☐

v. Collection of greater numbers.

☐

vi. Collection of Math teachers in your school.

☐

Q3. If $X = \{3, 4, 5, 6\}$ and $Y = \{e, f, g, h\}$, fill in the blanks with the symbols $\in$ or $\notin$.

i. 5 _____ X

iv. c _____ Y

ii. 8 _____ X

v. g _____ Y

iii. k _____ X

vi. 4 _____ Y

Q4. If $A = \{5, 6, 7, 8\}$ and $B = \{a, b, c, d\}$, fill in the blanks with the symbols $\in$ or $\notin$.

(i) 5 _____ A

(iv) c _____ B

(ii) 9 _____ A

(v) b _____ B

(iii) m _____ A

(vi) 7 _____ B

Q5. Write in tabular form.

i. Set of first seven natural numbers.

Solution:

ii. Set of the provinces of Pakistan.

Solution:

iii. Set of first ten natural numbers exactly divisible by 4.

Solution:

iv. Set of letters in the word "nation".

Solution:

v. Set of science teachers in your school.

Solution:

vi. Set of names of colours of national flag.

Solution:

Q6. If $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4, \dots, 10\}$ then which of the following statements are true or false.

- i. $1 \in B$ _____
- ii. $7 \notin A$ _____
- iii. $e \in A$ _____
- iv. $4 \notin B$ _____
- v. $7 \notin A$ _____
- vi. $c \in B$ _____
- vii. $6 \notin A$ _____
- viii. $d \in A$ _____

TYPES OF SETS

Finite sets:

A set which has limited number of elements is called finite set.

Example: set $A = \{2, 4, 6\}$ has limited number of elements, so it is finite set.

Infinite sets:

A set which has unlimited number of elements is called infinite set.

Example: set $B = \{1, 2, 3, 4, \dots\}$ has unlimited number of elements, so it is infinite set.

Empty/void/null sets:

A set which has no element is called an empty or null or void set. It is denoted by $\emptyset$ or $\{\}$.

Example:

- i. $C =$ Set of natural numbers less than 1
- ii. $D =$ Set of triangles with four sides

Singleton set:

A set having only one element is called singleton.

Example:

$$A = \{5\}, Q = \{4\} \text{ and } T = \{0\}$$

Equal sets:

When all the elements of two sets are the same, they are called equal sets.

Two sets A and B are called equal sets if they have all elements same. Symbolically we represent as $A = B$ and read as: Set A is equal to set B.

Example:

$$P = \{x, y, z\}, Q = \{y, z, x\}, \text{ so set } P = Q$$

Equivalent sets:

If two sets have same number of elements, they are called equivalent sets.

Two sets A and B are called equivalent sets if the number of elements of both sets are same.

Symbolically we write as: $A \sim B$ and read as: Set A is equivalent set to set B.

Example:

$$X = \{1, 2, 3, 4\}, Y = \{a, b, c, d\}, \text{ so set } X \sim Y.$$

Subset of a set:

If each element of set A is also an element of set B then set A is called subset of set B. Symbolically we write as $A \subseteq B$.

Example:

i. $A = \{1\} \subseteq B = \{1, 2, 3\}$

ii. $B = \{1, 2, 3\} \subseteq C = \{1, 2, 3, 4\}$

Superset of a set:

If A is subset of B then the set B is called superset of A.
Symbolically we write it as ' $B \supset A$ '.

Example:

- i. $B = \{1, 2, 3\} \supset A = \{1\}$
- ii. $D = \{4, 3, 2, 1\} \supset B = \{1, 2, 3\}$

Proper subsets of a set:

If set A is subset of set B and also A is not equal to B then "A is proper subset of set B. Symbolically we write as $A \subset B$.

Example:

$$B = \{1, 2, 3\} \subset C = \{4, 3, 2, 1\}$$

Improper subsets of a set:

If set A is subset of set B and A is also equal to B then "A is called improper subset of set B". Symbolically we write as $A \subseteq B$.

Example:

$C = \{1, 2, 3, 4\}$, $D = \{4, 3, 2, 1\}$, so set C is improper subset of set D.

EXERCISE 1.2

Q1. Give reasons why following collections are not set.

(i) $\{d, o, o, r\}$

(ii) $\{\star, O, \star, \}$

(iii) $\{a, f, d, a\}$

(iv) $\{2, 2, 3, 3, 4, 4\}$

Q2. Which of the following are finite or infinite sets?

i. $C = \{1, 2, 3, \dots, 1000\}$.

ii. $D = \{1/2, 1/3, 1/4, \dots\}$.

iii. Set of pages in a book.

iv. Set of all prime numbers.

v. Set of colours in a standard rainbow.

vi. Set of integers from -10 to 10.

Q3. Which of the following are empty sets?

(i) Set of months with 30 days in February.

(ii) Set of integers that are even.

(iii) Set of people who can fly without assistance.

(iv) Set of days in a week that are longer than 24 hours.

Q4. Which pairs of the following are equal sets?

(i) $\{a, b, c\}$ and $\{c, b, a\}$

(ii) $\{5, 10, 15\}$ and $\{10, 15, 5\}$

(iii) $\{x, y\}$ and $\{y, z\}$

(iv) $\{1, 2, 3, 4\}$ and $\{4, 3, 2, 1\}$

Q5. Which pairs of the following are equivalent sets?

(i) $\{a, b, c, d\}$ and $\{1, 2, 3, 4\}$

(ii) $\{x, y\}$ and $\{p, q, r\}$

(iii) $\{f, i, l, e\}$ and $\{p, i, l, e\}$

(iv) $\{5, 6\}$ and $\{9, 10, 11, 12, 13\}$

Q6. Which of the following statements are true or false?

i. $C = \{a, b, c, d\}$ if C is the set of vowels in the English alphabet.

ii. $A \subseteq B$ if $A = \{1, 3, 5\}$ and $B = \{1, 2, 3, 4, 5\}$. _____

iii. If $F = \{0, 1, 2\}$ and $G = \{2, 1, 0\}$, then $F = G$. _____

iv. If $P = \{2, 4, 6\}$ and $Q = \{6, 4, 2\}$, then $P = Q$. _____

v. If H is an empty set, then H is equivalent to any set with zero elements. _____

Q7. Choose appropriate symbol to indicate the relation between sets ($=, \subseteq, \not\subseteq, \supseteq, \sim, \subset$)

i. $\{2, 4, 6\}$ _____ $\{1, 2, 3, 4, 5, 6, 7\}$

ii. $\{3, 6, 9\}$ _____ Set of multiples of 3 less than 15.

iii. $\{\text{apple, banana, cherry}\}$ _____ $\{\text{cherry, banana, apple}\}$

iv. $\{x, y, z\}$ _____ $\{a, b, c, d\}$

v. $\{1, 2, 3\}$ _____ $\{2, 3\}$

vi. $\{a, b, c\}$ _____ $\{a, b, c, d, e\}$

vii. $\{5, 10, 15\}$ _____ $\{10, 20, 30\}$

Q8. Describe each set using finite, infinite or empty.

i. The set $\{2, 4, 6, 8, 10\}$ is an example of a _____ set.

ii. The set of all natural numbers $(1, 2, 3, \dots)$ is an example of an _____ set.

iii. The set with no elements, denoted as $\{\}$ or $\emptyset$, is called an _____ set.

iv. The set of days in a week $\{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$ is a _____ set.

v. The set of all points on a line is an example of an _____ set.